

Ganesh Gupta

817-822-1131 | iamgs10rk@gmail.com | linkedin.com/in/ganeshhgupta | github.com/ganeshhgupta

EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Engineered a pluggable distributed executor routing data pipeline nodes across Docker and AWS Lambda, backed by an S3 artifact store eliminating redundant recomputation at serious scale.
- Deployed a RAG data pipeline on AWS SageMaker for real-time decision support at sub-200ms p99 latency with RTO under 2 minutes.
- Replaced polling with SSE for live data streaming, cutting redundant API calls 90% across a system handling millions of daily transactions at 99%+ uptime.
- Owned backend services and APIs (Python/FastAPI) serving as the data access layer for downstream product and analytics consumers.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Rebuilt a retrieval data pipeline with hybrid dense-sparse indexing and re-ranking, improving recall 61% to 89% on a 2K-query eval set.
- Built a model observability framework tracking drift and latency regressions in production, enabling governance-aligned rollback workflows.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Cut nightly financial data aggregation blocking downstream settlement from 8 minutes to 90 seconds by redesigning SQL with indexed views and batch partitioning at scale.
- Built and operated 25+ production microservices in Java, C++, and SQL-backed distributed data systems for 1M+ users, sustaining 99.9% uptime over 3 years on AWS (EC2, RDS, SQS).
- Standardized failure-handling contracts across 25+ financial services, reducing production escalations 60% in one quarter via consistent retry, timeout, and propagation semantics.

PROJECTS

LangFetch

tinyurl.com/LangFetch

- Agent decomposing multi-table questions into dependency-ordered SQL subqueries with intermediate result passing, built for enterprise data queries at scale.

Distributed KV Store / VPC Control Plane Simulator

github.com/ganeshhgupta/distributed-kv-store

- Raft-consensus distributed KV store handling concurrent writes with linearizable reads and log compaction; companion VPC control plane simulator modeling subnet routing.

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- Full-stack data platform with FastAPI backend, React/TypeScript dashboard, and PostgreSQL, tracking drift and latency metrics across model versions.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages: Python, SQL, Java, C++, TypeScript

Data Engineering: Distributed Data Pipelines, Batch Partitioning, SQL Optimization, Data Aggregation at Scale, ETL

Data Stores: PostgreSQL, SQL/Relational Databases, NoSQL, S3

Cloud & DevOps: AWS (EC2, RDS, SQS, S3, Lambda, SageMaker), GCP (Cloud Run), Docker, Kubernetes, CI/CD

CERTIFICATIONS

AWS Certified Developer Associate | Google Cloud Professional Machine Learning Engineer